

Summative Report: Predicting Medical Insurance Charges

Results

My model tries to guess how much money a person pays for health insurance in one year, so my target variable is charges, which is the cost in US dollars. I used a dataset from Lantz (2013) with 1,338 people, and for each person it shows their age, BMI, number of children, sex, if they smoke, and the part of the US they live in. First I cleaned the data: there were no empty values, I deleted one row that was repeated, and I changed the text columns (sex, smoker and region) into 0 and 1 so the model could use them. Then I used 80% of the data to teach the model and kept 20% to test it on people it had not seen before, which shows if it can really make good guesses for new people. My first model got an R^2 of 0.81, an RMSE of about \$5,956 and an MAE of about \$4,177, and when I tried using the log of charges the model got worse, with an R^2 of 0.72. In my graphs I saw that smokers with a BMI over 30 paid a lot more money than other smokers, so I added this to the model, and I also added age squared because cost goes up faster for older people. My final model was the best, with an R^2 of 0.91 and an RMSE of about \$4,158, which means it explains about 91% of why charges are different between people, and its MAE of about \$2,394 means most guesses were only around two thousand dollars off. The most important thing I found is that smoking raises the cost the most, by about \$13,200 a year, and smokers who are also obese pay about \$19,600 more on top of that, while sex and region did not change the cost much, which I could also see in my box plots. My residual graphs showed that most guesses were close, with about 80% of them within \$2,000, but the model guessed too low for a few people with very high costs, maybe because of health problems that are not in the data. In the end, my model works well for most people and shows that smoking and being obese are the biggest reasons for high insurance costs, which could be useful for both insurance companies and customers, but adding health history would make it even better.

References

Lantz, B. (2013). *Machine learning with R*. Packt Publishing.

Stedy. (n.d.). *Machine-Learning-with-R-datasets* [Data set]. GitHub. Retrieved October 3, 2026, from <https://github.com/stedy/Machine-Learning-with-R-datasets>